Year 11 Admission Assessment v3.1: optional sciences, mark scheme

Version 3.1, October 2026. Use with `Y11/Y11_Optional_Sciences_v3.1` (the student paper: Biology, Chemistry and Physics, 15 marks and 15 minutes each). The core science paper (10 marks, everyone) is in the main student paper and its assessor instructions; this document covers only the optional sciences.

Reusable: print once and keep in the assessor's folder. Do not write on these pages. Record each candidate's marks on their own **grading sheet** (`Y11_Grading_Sheet_v3.1`): each science out of 15, the band, NT and L, and a "not taken" tick for any science not sat.

Who this is for. Students applying for **Year 11** (age 15 on 1 September), who join the **second year of a two-year IGCSE course**. A student from a Chinese mainstream school has usually **completed Grade 9 (初三)**: biology in Grades 7 and 8, physics in Grades 8 and 9 (including electricity and magnetism), and chemistry in Grade 9 only (atoms, the periodic table, acids and bases, simple equations and mass calculations by proportion; **the mole is not taught** before 高一). A student joining mid-year is partway through Grade 10 (高一). Ask which grade and textbooks the student has used.

What it measures. Each paper tests the **first year of IGCSE** in that science (biology: cells, enzymes, nutrition, transport in plants and humans, respiration and gas exchange; chemistry: particles, atomic structure, bonding, formulae and equations, the mole and reacting masses, acids, bases and salts, the periodic table; physics: motion and forces, energy, density and pressure, thermal physics, waves and light), with at least one calculation, one data question, one two-mark "explain" question and one working-scientifically question in each. The question is whether the student can **join the second year of that separate science, taught in English**, with a year's syllabus to catch up. Section 5 says how the result feeds that judgement.

1. How to give the optional sciences

Which sciences	Only the sciences the student will take in Year 11, as agreed with the family before the day (from the application form or the first conversation). The assessor ticks them in the box on the cover: **Biology, Chemistry, Physics**. If the choice is still open, give the sciences under consideration (all three take about 50 minutes with a break). Tick "not taken" on the grading sheet for any science not sat. Never give a science as a surprise.
When	After the cognitive ability screen, in the order Biology, Chemistry, Physics (leave out any not taken), with a **short break after two sciences**. If the day is too long, the optional sciences (and the interview) may be on a **second visit**: tell the family in advance.
Time	**15 minutes each**, timed strictly, with a two-minute warning. Do not run two sciences together as one 30-minute block: the time limit is part of the measure.
Materials	The student paper, a pencil, a rubber and a ruler. Keep a **scientific calculator** on your desk: give it for **Chemistry question 7** and for the **whole Physics section**, and take it back after. In a group, give each student theirs when they reach that point.
Marking	Mark in the boxes on the paper. B1 is a mark for a correct answer or point; M1 is a method mark (given for correct working even if the final answer is wrong); A1 is an accuracy mark for the final answer (usually needs the M1). Write **NT** beside any question the student says they have not been taught and **L** beside any question said once in Chinese.

Say (before the first science): "You will do [Biology, Chemistry and Physics / the sciences ticked on your cover]. Each one has 15 marks and takes 15 minutes. I can read any question to you in English, as many times as you like. Write all your answers in English. If you have not learnt a topic, tell me and go on to the next question. You can use a calculator only where the paper says so, and I will give it to you then. Do you have any questions?"

Say (at the start of each science): "Turn to the [Biology / Chemistry / Physics] page. You have 15 minutes. Look at every question: later questions may be on topics you know. Start now." Then at 13 minutes: "Two more minutes." At 15 minutes: "Stop now, please. Thank you."

Access rules (the pack's rules for maths and science)

	Rule
Reading aloud	Any question may be read aloud in English, as often as the student asks. You may point to the diagram, table or graph. Do not explain the science or define a word (for example "denatured" or "anomalous"): reading the question is all.
Chinese	Not used by the assessor. If a Chinese-speaking colleague is present and the student clearly cannot follow the English after it has been read aloud, the colleague may say the question **once** in Chinese. Write **L** in the margin. The mark still counts. Record the number of L questions on the grading sheet: it is evidence about English.
Answers	**All answers in English.** Numbers, symbols, formulae, ticks, circles, drawings and working all count. An explanation written in Chinese scores 0, including the two-mark "Explain" answers.
Calculator	Only where the paper says: **Chemistry question 7** and the **whole Physics section**. Biology: no calculator (the arithmetic is simple).
Not yet taught	If the student says they have not learnt a topic, write **NT** beside the question. It scores 0. List NT topics on the grading sheet; Section 5 says how to allow for them. The tables below say which questions are likely NT for a student from a Chinese school.
Stopping	The student may stop a science when they have done what they can. Encourage them to look at every question first. If a student is distressed, stop, take a break and record it.

Extra time, a reader or other arrangements for a known need are allowed: record them on the grading sheet.

2. Biology (15 minutes, 15 marks)

What a Year 11 entrant may not have been taught. Chinese middle-school biology (Grades 7 and 8) covers cells, plants, transport, digestion, breathing and the circulation well, so most of this paper is familiar. The **vocabulary of IGCSE biology in English** (chloroplast, magnification, xylem, transpiration, ventricle, denatured, active site, anomalous) is the main barrier: read questions aloud when asked. Likely NT: question 2(b) (Benedict's test), question 6(b) (denaturing and the active site, a 高一 idea).

Q	Topic	Answer	Marks and partial credit	May not have been taught?
1	Plant cell structure	(a) D (b) B	B1 each. The letter alone is enough; accept the letter with the correct name. (a) C (the nucleus) scores 0: the nucleus controls the cell but has no chlorophyll. (b) A (the cell wall) scores 0: the wall supports, the membrane controls what enters and leaves.	Secure (Grade 7 biology: 细胞壁, 细胞膜, 叶绿体).
2	Food tests: reading a table	(a) A (b) Food B contains (reducing) sugar	(a) B1: A only (starch with iodine and protein with biuret). (b) B1: reducing sugar, sugar or glucose. Do not accept "starch" or "protein".	The iodine test for starch is Grade 7. Benedict's and biuret tests are not usually taught before IGCSE, so (b) may be NT; (a) can be answered from the table if the student knows what iodine and biuret show.
3	Magnification (calculation)	600	M1 for 30 ÷ 0.05 seen or used; A1 for 600 (accept ×600). 0.0017 (0.05 ÷ 30) scores 0. A slip in the division (for example 60) scores M1 only. No calculator.	Chinese microscope work gives "magnification = eyepiece × objective"; the drawing-to-real calculation is IGCSE-specific, but the formula is printed, so the arithmetic (dividing by 0.05) is the test.

Q	Topic	Answer	Marks and partial credit	May not have been taught?
4	Transport in plants	(a) X (xylem) (b) transpiration	B1 each. (b) accept "transpiration" or a correct description ("water evaporates from the leaves"). Do not accept "respiration" or "photosynthesis".	Secure (Grade 7 biology: 导管 xylem, 筛管 phloem, 蒸腾作用 transpiration). The English word may be new: a correct description scores.
5	The heart	(a) D (b) B	B1 each. (a) C (left atrium) scores 0. (b) D (left ventricle, which pumps to the body) is the common confusion and scores 0. A student who answers A or C has reversed atria and ventricles: note it.	Secure (Grade 7 biology: four chambers, the two circulations). "Ventricle" (心室) in English may need the read-aloud; the diagram carries the question.
6	Enzymes: data and explain	(a) 40 (b) The enzyme is denatured; the active site changes shape so the substrate no longer fits	(a) B1: accept 39 to 41. (b) 2 marks: B1 "denatured" (accept "destroyed by the heat", "its shape has changed"); B1 the active site changes shape, or the substrate no longer fits or binds. "The enzyme dies" scores 0 for the first point (enzymes are not alive), but a correct second point still scores. "It is too hot" alone scores 0: it repeats the question.	Grade 7 biology meets digestive enzymes and that temperature affects them; "denatured" and "active site" are 高一 biology. (a) is secure; (b) may be NT, and a student who writes "the enzyme is destroyed by heat" without the active site shows the idea and scores 1.
7	Working scientifically: anomaly and mean (data)	(a) 45 circled (b) 30	(a) B1. (b) B1 for 30 ((29 + 31) ÷ 2). 35 (the mean of all three) scores 0 for (b) but (a) still scores if 45 is circled.	Means and odd results are familiar from maths; "anomalous" is explained in the question. Secure.

Band guide (provisional): 0 to 4: **1**; 5 to 7: **2**; 8 to 11: **3**; 12 to 15: **4**. Look at the NT questions before deciding (Section 5).

Not tested (not enough time): osmosis and diffusion, the digestive system in detail, the lungs and alveoli, the word equation for respiration. Ask about them in the interview if they matter.

What to look for

- **Explaining** (question 6b) separates a student who has learnt biology as explanation from one who has learnt it as facts: "denatured, active site changes shape, substrate cannot fit" in any words is IGCSE-ready thinking, even in simple English.
- **An English barrier** looks like: blank or wrong answers that become right when the question is read aloud, or correct biology in the wrong words (for example "the water goes out of the leaf as gas" for transpiration). Note the words the student did not know in the assessor's comment.
- **Strong responses** give units and precise values (40 °C, 30 breaths per minute, ×600) and handle the data questions (2, 6a, 7) quickly, leaving time for the explanation.

3. Chemistry (15 minutes, 15 marks)

What a Year 11 entrant may not have been taught. Chinese chemistry starts in **Grade 9 only**, so a Year 11 entrant has had one year: the particle model, atomic structure with electron-shell diagrams (原子结构示意图), the layout of the periodic table, acids, bases, pH and neutralisation, chemical equations, conservation of mass and **mass calculations by proportion** (化学方程式的计算). **The mole is not taught** before 高一, and nor are dot-and-cross diagrams, "covalent" as a term, or the group trends of the periodic table. Likely NT: questions 3(b), 4(a) and the mole method in 7(b), although 7(b) can be done by proportion. A student who comes mid-way through 高一 may know the mole.

Q	Topic	Answer	Marks and partial credit	May not have been taught?
1	Particles: states of matter	C	B1.	Secure (Grade 9 chemistry and Grade 8 physics: 分子, the particle model).
2	Atomic structure	(a) 19 (b) 7	(a) B1 for 19 (9 + 10). 9 (protons only) or 28 (all particles) scores 0. (b) B1: accept 7, VII, 17 (the IUPAC group number) or "the halogens".	Electron-shell diagrams with 2, 7 are Grade 9, so the diagram is familiar. "Mass number" (质量数) is 高一; Grade 9 uses relative atomic mass, so a student may answer 19 from that. The group from the outer electrons is touched on in Grade 9.
3	The periodic table	(a) Z (b) W	B1 each. (b) accept W only: X (Group 2) reacts slowly with water; Y and Z are non-metals.	The layout is Grade 9 (元素周期表), but the groups' names and reactivity trends are 高一: (b) may be NT. (a) is a fair test of whether the student knows where the noble gases sit.
4	Covalent bonding	(a) covalent (b) 4	(a) B1: covalent, covalent bonds, or "sharing (pairs of) electrons". "Ionic" scores 0. (b) B1: 4.	Dot-and-cross diagrams and the word "covalent" (共价键) are 高一: (a) may be NT. The diagram lets any student count the shared pairs for (b).
5	Acids, bases and salts	(a) D (b) sodium chloride	(a) B1. (b) B1: sodium chloride or NaCl. "Salt" alone scores 0 (too general).	Secure (Grade 9: pH, 中和反应, salts). Naming compounds in English is the likely barrier: accept the formula.
6	Working scientifically: variables	(a) The metal (which metal is used) (b) One control variable	(a) B1: the metal, the type of metal. "The number of bubbles" scores 0 (that is what she measures). (b) B1: the volume of acid; the concentration of acid; the temperature; the size, mass or surface area of the metal piece; the time (one minute); the type of acid. Do not accept "the metal" or "the bubbles".	Secure: 控制变量法 (controlling variables) is taught from Grade 8 physics, so the idea is familiar even if "variable" needs the read-aloud.
7	Reacting masses: explain and calculate (calculator allowed)	(a) Magnesium reacts with (combines with) oxygen from the air; the oxygen adds mass, so the product is heavier than the magnesium (b) 20 g	(a) 2 marks: B1 for magnesium reacting or combining with oxygen (from the air); B1 for the oxygen adding mass (the product contains oxygen atoms as well as magnesium; mass of product = mass of magnesium + mass of oxygen used). "It gets heavier because it burns" scores 0. (b) 2 marks: M1 for moles of Mg = 12 ÷ 24 = 0.5, or Mr(MgO) = 40 used with 0.5 mol, or the mass ratio 48 : 80 (or 24 : 40) set up; A1 for 20 g. The **proportion method** taught in China (24 g of Mg makes 40 g of MgO, so 12 g makes 20 g) earns full marks. 8 g (the mass of oxygen only) scores M1.	Conservation of mass (质量守恒定律) and mass calculations from equations are Grade 9, so most students can do both parts **without moles**. The mole itself is 高一: mark NT only if the student says the calculation is new to them.

Band guide (provisional): 0 to 4: **1**; 5 to 7: **2**; 8 to 11: **3**; 12 to 15: **4**. Allow for NT (Section 5): a student from a Chinese school may NT 3(b), 4(a) and part of 7 and still be ready.

Not tested (not enough time): ionic bonding and formulae of ions, separating mixtures, metals and reactivity in detail, percentage composition. Ask about them in the interview if they matter.

What to look for

- **Question 7** is the best guide: a student who explains the mass gain through oxygen combining, and calculates 20 g by any correct route, is ready for quantitative chemistry in IGCSE even with the mole still to learn.
- **An English barrier** looks like: chemistry correct in symbols (NaCl, the shells, 2 : 1 ratios) but the words missing or wrong (sodium chloride, covalent, alkaline). Record the words the student did not know.
- **Weak chemistry** (as opposed to NT) looks like: wrong answers on Grade 9 content that was taught: question 1, 2(a), 5 and 7(b) by proportion.

4. Physics (15 minutes, 15 marks)

What a Year 11 entrant may not have been taught. Chinese physics runs through Grades 8 and 9: speed, density, forces, pressure, simple machines, work and power, heat transfer, sound, light (reflection and refraction) and, in Grade 9, electricity and magnetism. **Acceleration and speed-time graphs, wavelength, frequency and wave speed, and the formula for gravitational potential energy are 高一** topics, so questions 3(b), 5 and 6(a) may be NT, although every formula needed is printed in the box at the start of the section. A student joining mid-way through 高一 usually knows motion graphs and acceleration.

Q	Topic	Answer	Marks and partial credit	May not have been taught?
1	Resultant force	20 N to the right	B1 for both the size and the direction (20 N alone, or "right" alone, scores 0). 40 N scores 0.	Secure (Grade 8 physics: 同一直线上二力的合成).
2	Density (calculation)	2.7 g/cm^3	M1 for the volume 10 × 5 × 2 = 100 cm^3 (seen or used); A1 for 2.7. Accept 2700 kg/m^3. 27 (270 ÷ 10) scores 0.	Secure (Grade 8 physics: 密度 = 质量 ÷ 体积). Fast and accurate work here is expected.
3	Speed-time graph: describe and acceleration	(a) Constant (steady) speed of 12 m/s (b) 3 m/s^2	(a) B1: constant speed, steady speed, not accelerating; accept "stays at 12 m/s". "Stopped" or "not moving" scores 0: this is the distance-time confusion. (b) M1 for 12 ÷ 4 (change in speed ÷ time); A1 for 3. 48 (12 × 4) scores 0; 24 (the distance travelled in the first 4 s, from the area) scores 0.	Speed-time graphs and acceleration (加速度) are 高一: both parts may be NT. The formula is printed, so a student who reads the graph correctly can still score; a student who answers "stopped" in (a) has read it as a distance-time graph.
4	Thermal physics: cooling curves (data)	(a) 50 (b) black	(a) B1: accept 49 to 51. (b) B1.	Reading the graph is secure. Black surfaces as better emitters and absorbers of radiation appear in some Grade 9 textbooks (热传递), otherwise IGCSE: (b) may be NT, but many students know it from everyday experience.
5	Waves: wavelength and wave speed (calculation)	(a) 2 (b) 10	(a) B1: 2 m. 4 (the whole length) or 1 (half a wave) scores 0. (b) B1 for 10 (5 × 2); allow follow-through from the student's (a) (for example 5 × 4 = 20 if (a) was 4).	Wavelength, frequency and wave speed = frequency × wavelength are 高一: likely NT. Sound as a vibration is Grade 8, without the equation. The formula is printed; the test is whether the student can read a wavelength off a diagram.

Q	Topic	Answer	Marks and partial credit	May not have been taught?
6	Energy: GPE and the explain question	(a) 800 J (b) Friction (or air resistance) acts on the skateboard; energy is transferred to the thermal store (heats the wheels, ramp and air) and sound, so less is left as kinetic energy	(a) B1 for 800 (40 × 10 × 2). 80 (g forgotten) scores 0. (b) 2 marks: B1 for friction between the wheels and the ramp, or air resistance; B1 for energy transferred to the thermal store, heating, or sound (accept "wasted as heat", "dissipated"). "Energy is lost" alone scores 0: it must say where the energy goes.	Grade 9 physics covers gravitational potential energy and friction qualitatively; the formula mass × g × height is 高一 but is printed. (a) may be NT; (b) is within reach of any student who has met friction and heating.
7	Working scientifically: light, concluding and accuracy	(a) The angle of reflection equals the angle of incidence (b) One improvement	(a) B1: "r = i", "the angles are equal", "the angle of reflection is the same as the angle of incidence". "They both increase" scores 0 (a description, not the conclusion). (b) B1: repeat each angle and take a mean; use a narrower ray (a thinner slit); mark the ray with two points far apart before drawing it; use a sharp pencil or a larger protractor; make sure the mirror stands exactly on the drawn line; measure from the normal carefully. "Be more careful" or "use a better protractor" alone scores 0.	Secure: the law of reflection (光的反射定律) is Grade 8 physics, and investigations with a discussion of error are familiar from 探究 lessons.

Band guide (provisional): 0 to 4: **1**; 5 to 7: **2**; 8 to 11: **3**; 12 to 15: **4**. Allow for NT (Section 5): a student from a Chinese school may NT 3, 5 and 6(a) and still be ready.

Not tested (not enough time): pressure in liquids, work and power, specific heat capacity, refraction and lenses, the electromagnetic spectrum. Electricity and magnetism are second-year IGCSE topics and are not tested; a Chinese student will usually have done them in Grade 9.

What to look for

- **Using a printed formula on a new topic** (questions 3b, 5b, 6a) is the best guide to physics readiness: a student who has never met acceleration but reads the graph, chooses the right formula and gets 3 m/s^2 will manage the second year of IGCSE with the first-year topics taught quickly.
- **An English barrier** looks like: calculations right, descriptions and explanations blank or in the wrong words (for example "the speed is same" for constant speed). "Emitter", "constant", "resultant" and "anomalous" are the words most often unknown: record them.
- **Strong responses** show working, give units (g/cm^3, m/s^2, J) and in question 6(b) say where the energy goes, not just that it is "lost".

5. Band guide and the readiness judgement for separate sciences

Band guide for each science (provisional, review after the first 20 to 30 candidates):

Marks out of 15	Band	What it usually means
0 to 4	**1** Well below expected	Not ready to join this science a year in unless NT or English explains most of the gap.
5 to 7	**2** Working towards	Some of the first year secure; a catch-up plan for the rest is needed.
8 to 11	**3** Expected	Ready to join the second year of this science with the NT topics taught early in the term.
12 to 15	**4** Greater depth	Secure first-year content; likely to do well in the separate science.

Allow for NT and L before you read the band. The likely NT questions for a student from a Chinese school are: biology 2(b) and 6(b); chemistry 3(b), 4(a) and 7 (the mole method only); physics 3, 5 and 6(a). Add up the marks available on the questions that were **not** NT and judge the band on those (for example 9 out of 11 on the non-NT questions in physics is band 3 or 4, not band 2). Then list the NT topics in the assessor's comment: they are the catch-up plan. Treat a band 1 that NT or L explains as band 2 (as in the guide, Section 9).

How the result feeds the readiness judgement. A Year 11 entrant joins courses that are already a year in. For each separate science the student wants to take, the question is whether they can **catch up the first-year syllabus while following the second year in English**. Read the science band with the English band (reading and writing above all, because IGCSE science papers are long and the command words matter) and with the core science question on working scientifically. **This is guidance for the school to confirm** against its own syllabuses, timetable and support; it is not a set of fixed rules.

Result in a science	What it suggests	A possible plan (for the school to confirm)
Band 4, or band 3 with the gaps all NT; English D or E	Ready to join the second year of this separate science.	Separate science; the NT topics taught in the first weeks (or by a short bridging course before the term starts).
Band 3 or 4; English C	The science is secure; the English of science is the risk.	Separate science with support for subject language (key vocabulary, command words, writing explanations), and not too many essay-based subjects. Review after one term.
Band 2 with NT explaining most of the gap	Likely to manage with a structured catch-up.	Separate science with a written catch-up plan for the first-year topics (the NT list) and a review at half term; consider a bridging course.
Band 2 with errors on taught topics (the "Secure" rows above)	The first year of this science is not secure, and the second year builds on it.	Combined or co-ordinated science (if the school offers it), or fewer separate sciences, or a **Year 10 start** if the student's age allows; discuss with the family.
Band 1 not explained by NT or English	Joining this science a year in is not advisable.	Not this separate science in Year 11; a Year 10 start or a different subject mix. If the band 1 is across all three sciences with English D or E, further assessment before a decision (`00_Admissions_Guide.md`, Section 9).
Strong in two sciences, band 1 or 2 in the third	Usual profile: strength and interest are uneven.	Offer the two strong sciences; take the third only with a catch-up plan and the student's own commitment (check in the interview).
Good bands, but English B	The science can be followed with pictures and formulae; the exam papers cannot yet be read or written.	Only with an ESL course, a reduced subject load and close review, if the school's policy allows; otherwise discuss a preparatory route or a Year 10 start.

Record the readiness judgement for the science route (separate sciences, combined, or fewer subjects) on the grading sheet's readiness box and in the final mark sheet (`forms/Candidate_Summary_and_Decision`), with the NT topics named as the catch-up plan.

6. Recording the results

1. **Mark each science on the paper.** Every question has a mark box; write NT and L in the margin where they apply.
2. **Copy each total onto the grading sheet** (`Y11_Grading_Sheet_v3.1`): Biology / 15, Chemistry / 15, Physics / 15, each with its band (1 to 4), the NT topics, the number of L questions, and a tick in "not taken" for any science not sat.
3. **Write the assessor's comment**: the NT topics for each science (the catch-up plan), the English words that were a barrier, what the working-scientifically and explain questions showed, and whether the calculator was used.
4. **Judge readiness** for the science route with Section 5, together with the English bands and the core science result, and tick it on the grading sheet's readiness box.
5. **Complete the final mark sheet** (`forms/Candidate_Summary_and_Decision`) with the other results within two working days, and **file the marked paper** with the student's papers (or shred it under the school's policy).